

Chapter 86

3D Camera Tracking

This chapter presents an overview of using the Camera Tracker node and the workflow it involves. Camera tracking is used to create a virtual camera in Fusion's 3D environment based on the movement of a live-action camera in a clip. You can then use the virtual camera to composite 3D models, text, or 2D images into a live-action clip that has a moving camera.

For more information on other types of tracking in Fusion, see *Chapter 82, "Using the Tracker Node,"* in the DaVinci Resolve Reference Manual or *Chapter 22* in the Fusion Reference Manual.

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Introduction to Tracking

Tracking is one of the most useful and essential techniques available to a compositor. It can roughly be defined as the creation of a motion path from analyzing a specific area in a clip over time. Fusion provides a variety of different tracking nodes that let you analyze different kinds of motion.

Each tracker type has its own chapter in this manual. This chapter covers the tracking techniques with the Camera Tracker node.

What Is 3D Camera Tracking?

Camera tracking is used for match moving, and it's a vital link between 2D scenes and 3D scenes, allowing compositors to integrate 3D CGI elements into live-action clips. The Camera Tracker node calculates the path of a live-action camera and generates a virtual camera in 3D space. This virtual camera is intended to be identical to the actual camera that shot the scene, not only in terms of motion but in matching the lens focal length as well. The calculated position and movement of the virtual camera is central to realistically compositing 3D elements with live action.



An example of 3D elements integrated in a live-action scene

How Camera Tracking Works

Camera tracking begins by tracking the movement of fixed features from one frame to the next. To put it another way, camera tracking algorithms follow features that are “nailed to the set.” Objects in the scene that move independently of the camera movement in the shot, such as cars driving or people walking, cause poor tracks, so masks can be used to restrict the features that are tracked in order to improve the results. Additionally, it is helpful to provide specific camera metadata, such as the sensor size and the focal length of the lens. This information guides the scene reconstruction calculation, called a solver, toward generating a more accurate virtual camera.

The Camera Tracker’s purpose is to create a 3D animated camera and point cloud of the scene. A point cloud is a large group of points generated by the solver that roughly recreates the 3D positions of the tracked features in a scene. The point cloud can then be used as a guide when integrating other 2D or 3D elements alongside live-action features.

The Camera Tracking Workflow

Camera tracking has two main phases:

- 1 *Tracking*, which is the analysis of a scene.
- 2 *Solving*, which calculates the virtual 3D scene.

Once you complete these steps, an animated camera and point cloud are exported from the Inspector into a 3D composite. The Camera Tracker encompasses this complete workflow within one tool. Five tabs at the top of the Inspector are roughly laid out in the order in which you’ll use them. These five tabs are:

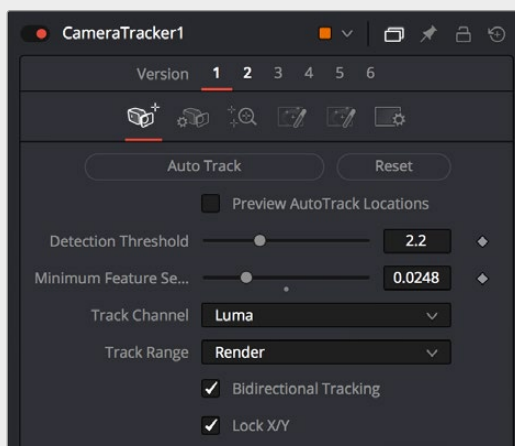
Track: Used to track a clip.

Camera: Configures the basic Camera parameters.

Solve: Calculates the 3D placement of the 2D tracking points and reconstructs the camera.

Export: Generates a Camera 3D node, a Point Cloud node, and a 3D scene in the node tree.

Options: Used to customize the look of the onscreen overlays.



The Camera Tracker tab

Clips That Don't Work Well for Camera Tracking

Even though the Camera Tracker is somewhat automatic, it sometimes needs your help. If you can identify potential issues before you even track or solve the shot, you can save yourself much time. Certain types of clips will cause more significant problems for camera tracking than others. Some are fixable, while for others you just have to admit defeat and figure out another solution. Here is a list of the types of shots to look out for, as they can be big headaches for camera tracking:

- **Lack of depth:** Camera tracking requires parallax in a clip in order to work. You must be able to identify objects further away and objects that are nearer as the camera moves. If everything is at the same distance from the camera, there is no way to calculate depth. In this case, it's better to skip the Camera Tracker node and find another solution.
- **Locked-off shots:** If the camera does not move, there is no way to calculate which objects are closer and which are nearer. Again, don't spend too much time in this situation; it is better to skip the Camera Tracker node and find another solution.
- **Tripod pans:** Similar to a locked-off shot, there is no way to calculate which objects are closer and which are nearer from a pan that remains centered on a locked-off tripod. Skip the Camera Tracker node and find another solution.
- **No detail:** Clips like green screens without tracking markers lack enough detail to track. If you are lucky enough to be involved in the shooting of these types of shots, including tracker markers makes it much easier to get a good track. Without detail, camera tracking will fail and you will need to find a more manual solution.
- **Motion blur:** Fast camera motion or slow shutter speeds can introduce motion blur, which will make it difficult to find patterns to track. It's worth trying shots like these to see if there are enough details to get a good solve, but know when give up and turn to another solution.
- **Rolling shutter:** CMOS-based cameras sometimes introduce distortion due to the shutter capturing different lines at slightly different times. This distortion can create significant problems for camera tracking. Sometimes it is possible to create motion vectors with the Optical Flow node to create new in-between frames without the wobble distortion of the rolling shutter. Then you can use the corrected image to connect to the Camera Tracker.
- **Parallax issues:** When objects at different distances in a shot overlap in the frame, the overlapping area can be misinterpreted as a corner. Having a tracker assigned to an overlapping angle like this will cause errors as the parallax starts to shift and the overlapping area slides. This can be solved in Fusion by removing that tracker before running the solver.
- **Moving objects:** It's difficult to capture a shot where objects in the clip do not move. People, cars, animals, or other object may move in and out of a shot. These objects move independent of the camera movement and must be eliminated or they will cause solving errors. You can fix these issues by masking out objects that are not "nailed to the set." The masks are then connected to the Track Mask input on the Camera Tracker node.

TIP: Some shots that cannot be tracked using Fusion's Camera Tracker can be performed in dedicated 3D camera-tracking software like 3D Equalizer and PF Track. Camera tracking data from these applications can then be imported in the Camera3D node in Fusion.

Outputting from the Camera Tracker

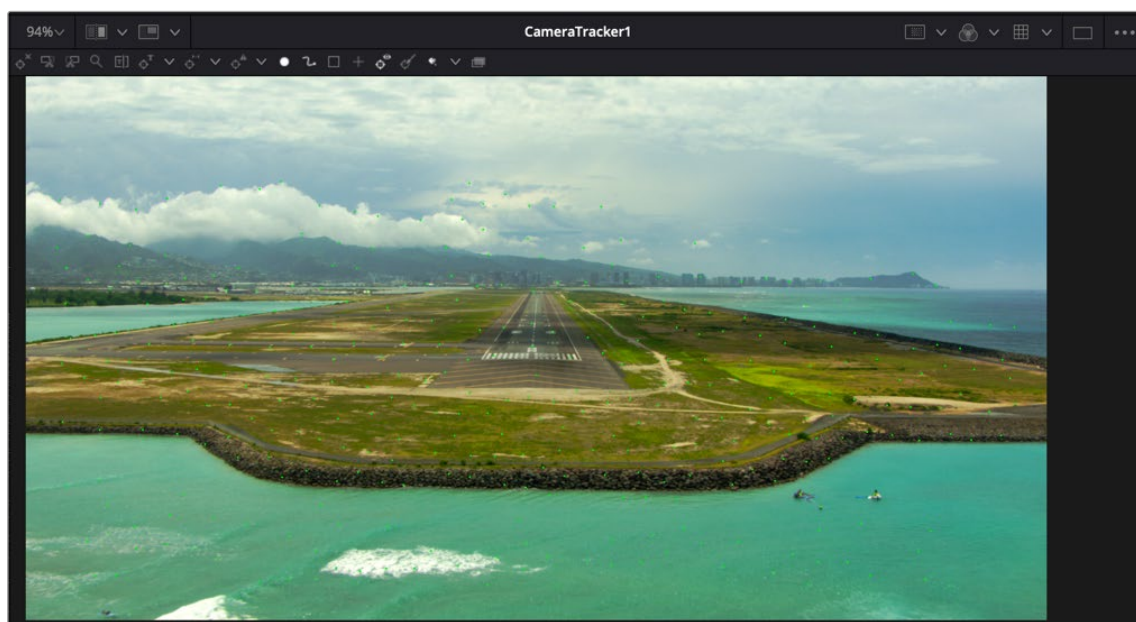
Unlike most Fusion nodes, the Camera Tracker node has two outputs:

- The primary output is a 2D view used when you are setting up the Track, refining the camera, and performing your initial solve.
- There is also a 3D output used after your initial solve for viewing the camera path and point cloud in 3D space. This view can be helpful when you are refining tracks to increase the accuracy of the solve and aligning your ground plane. It can be used simultaneously with the 2D output in side-by-side views.

Note that the selection of tracks in the 2D view and their corresponding locators (in the point cloud) in the 3D view are synchronized. There are also viewer menus available in both the 2D and 3D views to give quick control of the functionality of this tool.

2D View

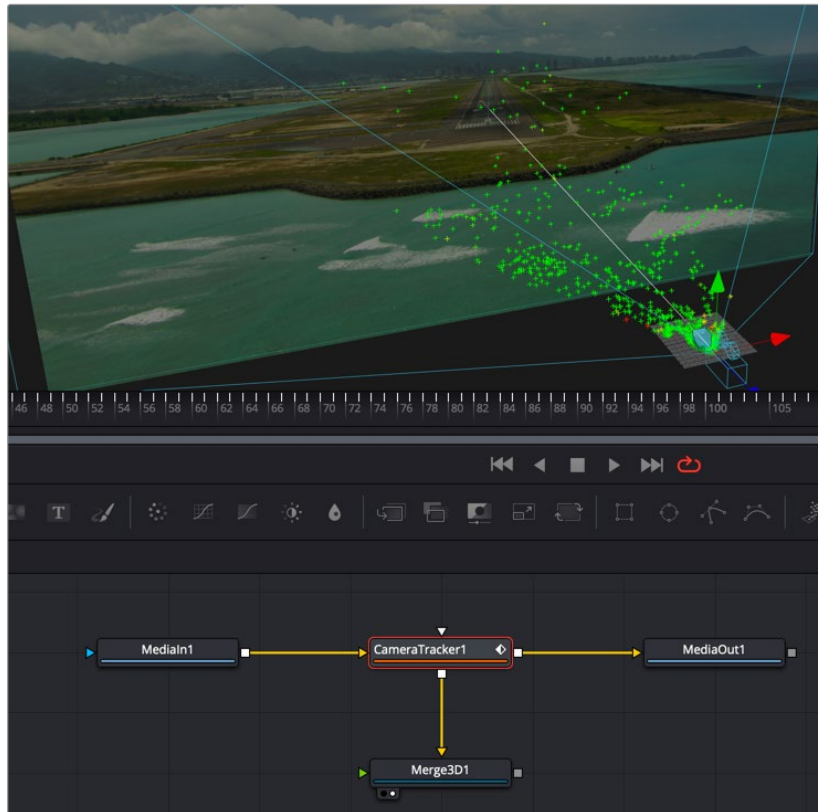
The 2D view is the primary display for the node. Viewing the node displays the image being tracked as well as overlay tracker markers and their motion paths. A dedicated toolbar gives you access to the common features used to track and solve a clip.



The Camera Tracker 2D output with toolbar and auto-track points

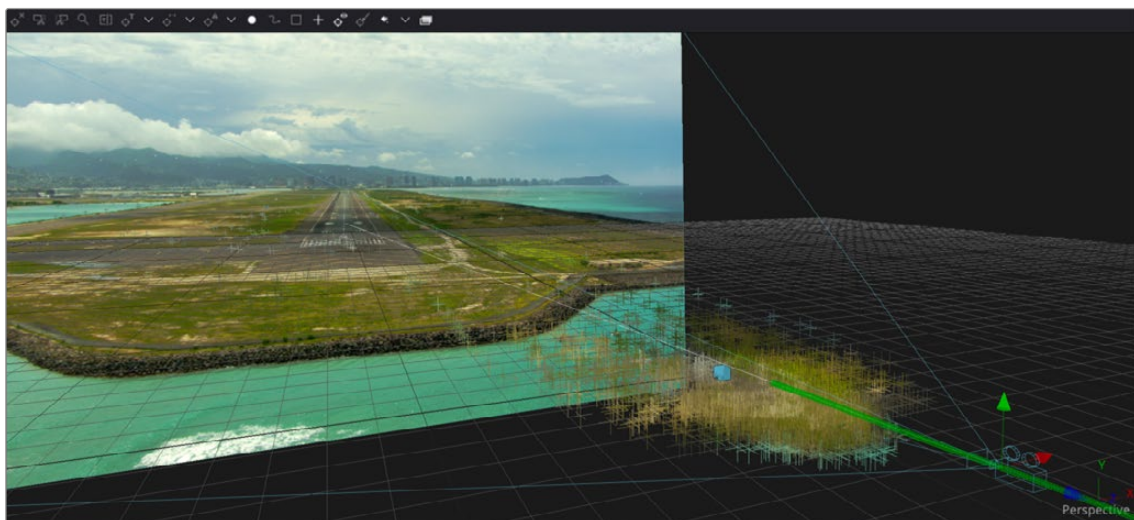
3D View

The second output of the Camera Tracker node displays a 3D scene. To view this, connect this 3D output to a 3D Transform or Merge 3D node and view that tool.



After solving, connecting the second output of the Camera Tracker node to a Merge 3D displays the point cloud in 3D

After an initial solve, the 3D output displays the point cloud and the camera, along with the image connected to it. Selecting points displays the Camera Tracker toolbar above the viewer, which gives control of various functions, such as renaming, deleting, and changing the colors of points in the point cloud.



3D output of a point cloud and a solved camera path

Auto-Tracking in the Camera Tracker

Tracking is the term used to describe the task of observing or analyzing a sequence of frames (or clip). The Camera Tracker node must take into account the movement of the source footage before it can determine the location and movement of the virtual camera. The Camera Tracker tool automatically searches for features that are high-contrast patterns within the clip and assigns trackers to those features. Having a wide distribution of tracking points across the frame and having points with long durations results in the best track.

Increasing Auto-Generated Tracking Points

Unlike the Tracker node, setting tracking points is entirely automatic in the Camera Tracker, but the Detection Threshold and Minimum Feature Separation sliders let you adjust the criteria by which tracking points are found. Lowering these parameters lets you increase the number of tracking points that will be found. This can be useful if the scene has too few points, which will prevent the solver from generating an accurate camera and point cloud. However, make these adjustments with care, since adding too many points may generate redundant trackers that slow down the entire process with minimal benefit.

Previewing Tracking Points

You can view the tracking points currently generated on the clip by viewing the Camera Tracker node and turning on the Preview AutoTrack Locations checkbox. This causes green tracking points to be displayed in the viewer as you play through the clip. Using this preview, you can decide whether you need to make adjustments to the Detection Threshold or Minimum Feature Separation to increase or decrease the number of tracking points that are found automatically.



Green tracker marks are added automatically to the features in an image

Bi-Directional Tracking

When performing a track, you can enable the Bidirectional Tracking checkbox, which first tracks forward from the start of the clip, and then tracks a second pass in reverse. This two-pass approach can potentially extend the duration of any given point by re-analyzing points initially identified in the forward pass. There is very little reason not to have this enabled unless you are very short on time. Bidirectional tracking takes longer, but it's usually worth it, and the process is reasonably quick considering the benefit.

Choosing a Tracking Algorithm

There are three available choices for the algorithm to use when tracking. The three options in the New Track Defaults section of the Inspector include:

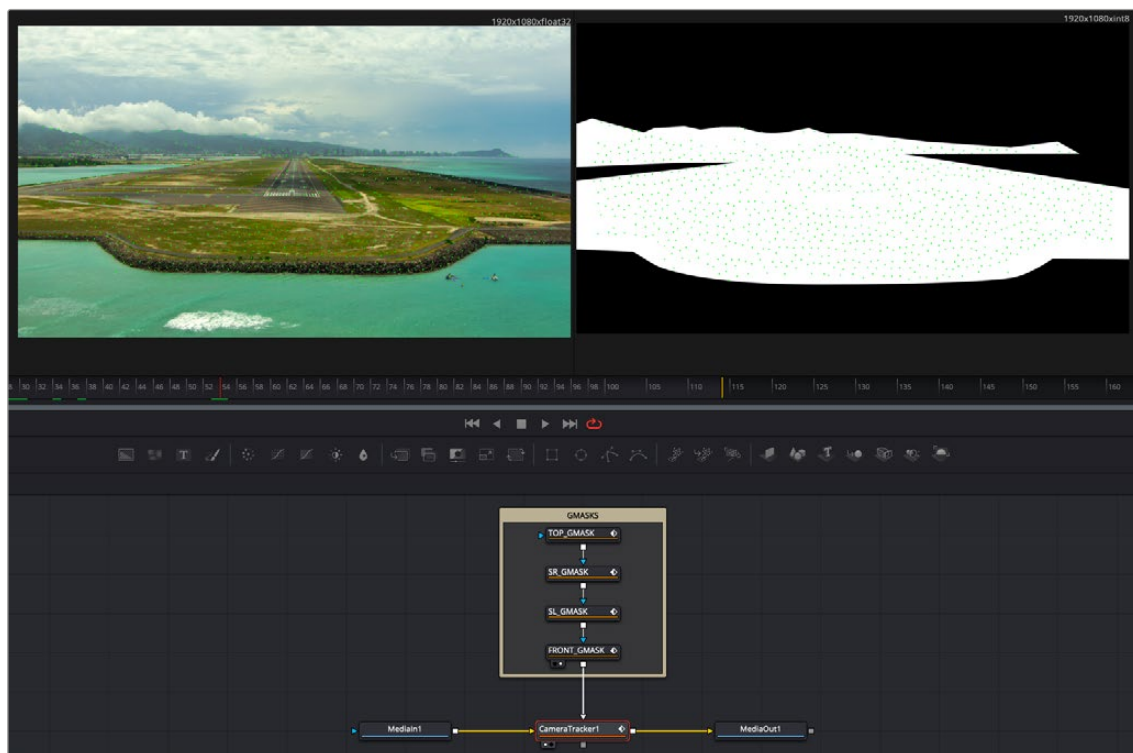
- **Optical Flow:** Usually your best choice, unless you have a great deal of criss-crossing objects in a clip.
- **Tracker:** A good second choice when Optical Flow can't be used due to motion estimation errors like criss-crossing objects.
- **Planar:** Mostly used in simpler clips, where the majority of the image consists of planar surfaces such as the facades of buildings.

Masking Out Objects

When tracking a clip, the Camera Tracker automatically generates trackers on feature details. However, not all features that stand out in a clip are appropriate for camera tracking. You only want to track features that are “nailed to the set.” In other words, objects that move independently of the camera motion, like moving cars and people, cause inaccuracies when camera tracking. You must eliminate these types of objects from the analysis.

The primary way of avoiding these problem areas is by masking. You connect a mask to the Camera Tracker node's Track Mask input to identify areas of a scene that the Camera Tracker can analyze. For example, if you have a clip of an airport runway along a shoreline, the waves of the water and moving clouds in the sky must be masked since they move independently of the camera.

When creating a mask, the fixed areas of the image to be analyzed for tracking should be encompassed in the white portion of the mask. All moving objects that need to be ignored should be encompassed in the black portion. The mask should then be attached to the Camera Tracker Track Mask input.



Masks used to omit the moving clouds and waves from being tracked by the Camera Tracker

By doing this, the tracker ignores the waves of the water and moving clouds. Unlike drawing a mask for an effect, the mask in this case does not have to be perfect. You are just trying to identify the rough area to occlude from the tracking analysis.



The original image to be tracked (left), and the occlusion mask of the clouds and water (right)

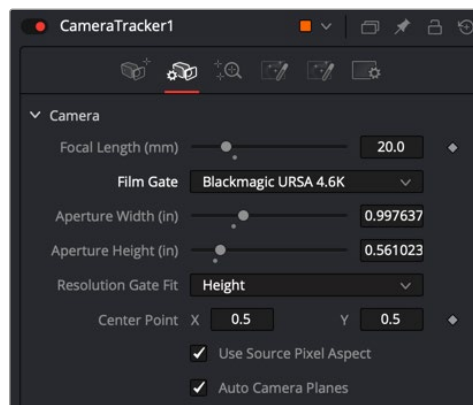
TIP: If there's a lot of motion in a shot, you can use the Tracker or Planar Tracker nodes to make your occlusion mask follow the area you want to track. Just remember that, after using the PlanarTracker or PlanarTransform node to transform your mask, you need to use a Bitmap node to turn it back into a mask that can be connected to the Camera Tracker node's Track Mask input.

Matching the Live-Action Camera

Once you have completed tracking, the next stage of this workflow requires the controls in the Camera tab. This is where you define the actual camera used on set, primarily the film gate size and focal length. This information should have been logged on the set to make available for post-production. When using camera-original media, you can sometimes locate this information in the file metadata.

To locate camera metadata, do the following:

- If you are using DaVinci Resolve, select the MediaIn node with the camera clip, open the Metadata Editor, and view the Camera metadata preset.
- If you are using Fusion Studio, display the metadata subview from the viewer toolbar.



The Camera tab in the Camera Tracker tool

If the actual values are not known, try a best guess. The solver attempts to find a camera near these parameters, and it helps the solver by giving parameters as close to the live action as possible. The more accurate the information you provide, the more accurate the solver calculation. At a minimum, try to at least choose the correct camera model from the Film Gate menu. If the film gate is incorrect, the chances that the Camera Tracker correctly calculates the lens focal length become very low.

Unlike the Track and Solve tabs, the Camera tab does not include a button at the top of the Inspector that executes the process. There is no process to perform on the Camera tab once you configure the camera settings. After you set the camera settings to match the live-action camera, you move to the Solve tab.

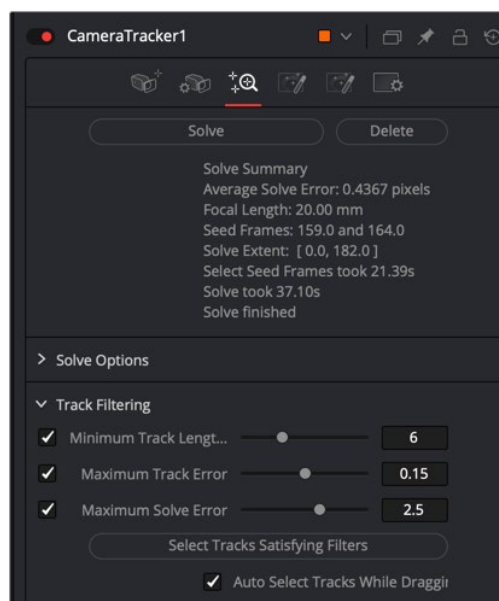
Running the Solver

The next step in this workflow involves the controls found in the Solve tab. Solving is a compute-intensive process in which the Camera Tracker analyzes the currently existing tracks to create a 3D scene. It generates a virtual camera that matches the live action and a point cloud consisting of 3D locators that recreate the tracked features in 3D space. The analysis is based on parallax in the frame, which is the perception that features closer to the camera move quicker than features further away. This is much like when you look out the side window of a car and can see objects in the distance move more slowly than items near the roadside.

The trackers found in the Track phase of this workflow have a great deal to do with the success or failure of the solver, making it critical to deliver the best set of tracking points from the very start. Although the masking you create to occlude objects from being tracked helps to omit problematic tracking points, you almost always need to further filter and delete poor quality tracks in the Solver tab. That's why, from a user's point of view, solving should be thought of as an iterative process.

To solve a camera's motion:

- 1 Click the Solve button to run the solver.
- 2 Filter out and delete poor tracks.
- 3 Rerun the solver.



The Solver tab after it has run and produced an average solve error of 0.4367 pixels

How Do You Know When to Stop?

At the end of the solve process, an Average Solve Error (sometimes called a reprojection error) appears at the top of the Inspector. This is the crucial value that tells you how well the calculation has gone. A good Average Solve Error for HD content is below 1.0.

You can interpret a value of 1.0 as a pixel offset; at any given time, the track could be offset by 1 pixel. The higher the resolution, the lower the solve error should be. If you are working with 4K material, your goal should be to achieve a solve error below 0.5.

Tips for Solving Camera Motion

When solving camera movement, it's important to provide accurate live-action camera information, such as focal length and film gate size, which can significantly improve the accuracy of the camera solve. For example, if the provided focal length is too far away from the correct physical value, the solver can fail to converge, resulting in a useless solution.

Additionally, for the solver to accurately triangulate and reconstruct the camera and point cloud, it is important to have:

- A good balance of tracks across objects at different depths, with not too many tracks in the distant background or sky (these do not provide any additional perspective information to the solver).
- Tracks distributed evenly over the image and not highly clustered on a few objects or one side of the image.
- The track starts and ends staggered over time, with not too many tracks ending on the same frame.

Using Seed Frames

The solver works by first constructing a partial solution between two seed frames. These seed frames are selected automatically. However, automatic selection adds time to the process. The time is reported in the solve summary at the top of the Inspector once you run the solver. You can select your own seed frames to speed the process and potentially get a better solve on trickier clips. The solver uses these seed frames to create an intermediate solution and then extends this forward and backward for the duration of the clip.

Selecting appropriate seed frames is not necessarily recommended unless you have some experience with camera tracking. Keeping the default Auto Select Seed Frames checkbox enabled in the Solve Options section of the Solver tab selects the best frames in most cases. However, you can disable the checkbox and use the Seed Frame 1 and Seed Frame 2 slider to select frames you believe achieve better results.

When choosing seed frames, it is important to satisfy two main characteristics:

- Seed frames should have lots of tracks in common.
- Seed frames should be far apart in perspective.

Sometimes There's Nothing You Can Do

Some shots that do not have enough camera motion to triangulate feature locations cannot be reconstructed with any useful accuracy. Ensuring that a shot is camera-trackable begins on set, with proper usage of track markers, and by ensuring that camera moves have enough perspective shifts for the solver to glean useful data from.

Cleaning Up Camera Solves

Sometimes the first solve will be good enough. Other times, it may take a few hours of cleaning up tracks to get a good solve, and sometimes it is impossible. With experience, one gets a feel for which tracks should be deleted and which should be kept, and which shots will be easy, difficult, or impossible to solve.

Be aware that deleting too many tracks can cause the Average Solve Error to increase, as the solver has too little information to work with. In particular, if there are fewer than eight tracks on any frame, mathematically there is not enough information to solve the clip. However, it is strongly recommended to use a lot more than eight tracks to get a robust and accurate solve.

IMPORTANT: If you are not familiar with camera tracking, it may be tempting to try to directly edit the resulting 3D splines in the Spline Editor in order to improve a solved camera's motion path. This option should be used as an absolute last resort. It's preferable, instead, to modify the 2D tracks being fed into the solver.

How to Judge Track Accuracy

The automatic color coding of tracking markers makes deleting false or poor tracks easier. After the solver runs, each tracker is assigned a solve error color that indicates which 3D locators match their 2D tracking points well, and which match up poorly.

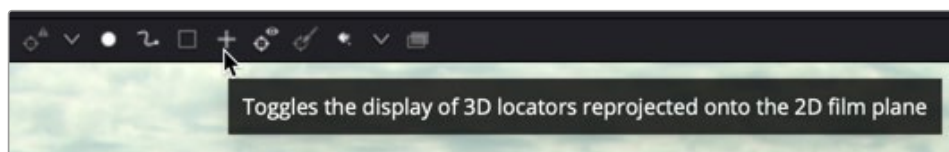
Green: Good. Tracked very well.

Yellow: Moderate confidence. Usually an acceptable track.

Orange: Low Confidence. May be acceptable in some situations.

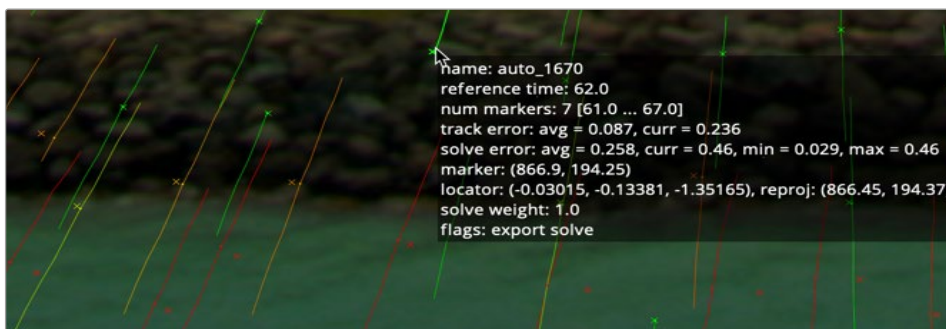
Red: No Confidence. The tracks have not solved well.

Hovering the pointer over any tracking point displays a large metadata tooltip that includes the solve error for the point. For a more visual representation of the accuracy, you can enable the display of 3D locators in the viewer by clicking the Reprojection Locators button in the viewer toolbar.



After a solve, the Camera Tracker toolbar can display Reprojection locators

When the tracking points are converted into a point cloud by the solver, it creates 3D reprojection locators for each tracking point. These Reprojection locators appear as small X marks near the corresponding tracking point. The more the two objects overlap, the lower the solve error.



Reprojection locators displayed with tracking points, and tooltip

The goal when filtering the trackers is to remove all red tracker marks and keep all the green marks. Whether you decide to keep both the yellow and orange or just the yellow is more a question of how many marks you have in the clip. You produce a better solve if you retain only the yellow marks; however, if you do not have enough marks to calculate the 3D scene, you will have to keep some of the better orange marks as well.

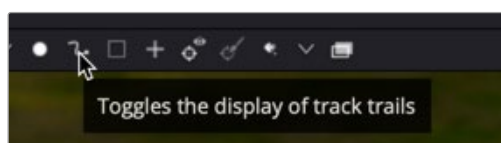
Tips for What to Keep and What to Delete

Understanding what false tracks look like, and then manually cleaning the track data to reduce it to a precise set of clear tracks, will result in a more accurate solve. When cleaning up any track—particularly yellow and orange color coded tracks—keep the following in mind:

- Keep all tracks with motion that's completely determined by the motion of the live-action camera.
- Delete tracks on moving objects or people and tracks that have parallax issues.
- Delete tracks that are reflected in windows or water.
- Delete tracks of highlights that move over a surface.
- Delete tracks that do not do a good job of following a feature.
- Delete tracks that follow false corners created by the superposition of foreground and background layers.
- Consider deleting tracks that correspond to locators that the solver has reconstructed at an incorrect Z-depth.

Deleting Tracks

You can manually delete tracks in the viewer or use filters to select groups of tracks. When deleting tracks in the viewer, it is best to modify the viewer just a bit to see the tracks more clearly. From the Camera Tracker toolbar above the viewer, clicking the Track Trails button hides the trails of the tracking points. This cleans up the viewer to show points only, making it easier to make selections. At the right end of the toolbar, clicking the Darken Image button slightly darkens the image, again making the points stand out a bit more in the viewer.



Hiding trails and darkening the viewer can make it easier to see and select poor tracks

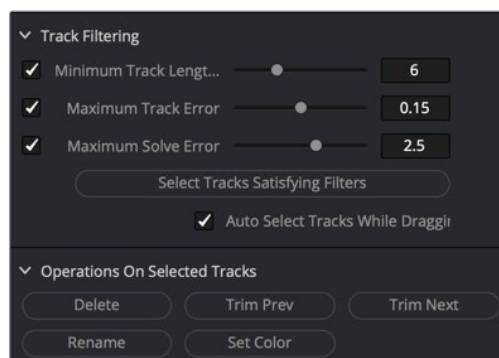
To begin deleting poor-quality tracks, you can drag a selection box around a group of tracks you want to remove and then either click the Delete Tracks button in the Camera Tracker toolbar or press Command-Delete.

You can hold down the Command key to select discontinuous tracking marks that are not near each other. If you accidentally select tracks you want to keep, continue holding the Command key and drag over the selected tracks to deselect them.

When deleting tracks, take note of the current Average Solve Error at the top of the Inspector and then rerun the solver. It is better to delete small groups of tracks and then rerun the solver than to delete one or two large sections. As mentioned previously, deleting too many tracks can have adverse effects and increase the Average Solve Error.

Using Filters to Delete Problem Tracks

The Solve tab includes filters that can be used to select groups of similar tracks by track length, track error, and solve error. These can be used to quickly select and delete poorly performing tracks that may be misleading to the resulting camera, leaving a concise list of accurate tracks.



Tracks can be selected using filters and deleted using the Operations On Selected Tracks buttons

For instance, it is generally best to run the solver using tracks with longer durations. Since shorter tracks tend to be less accurate when calculating the camera, you can remove them using the Filter section in the Inspector.

Increasing the Minimum Track Length parameter sets a threshold that each tracker must meet. Tracks falling below the threshold appear red. You can then click the Select Tracks Satisfying Filters button to select the shorter tracks and click Delete from the Options section in the Inspector.

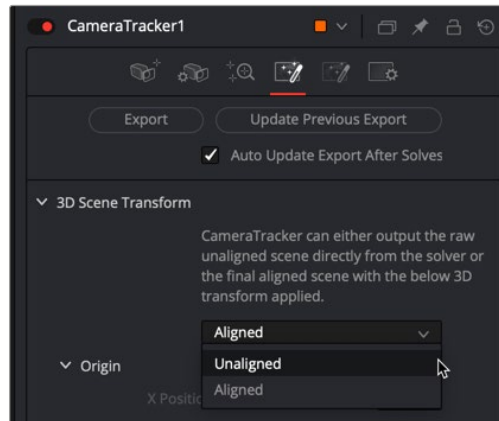
Exporting a 3D Scene for Efficiency

The Camera Tracker saves all its 2D tracks into the composition, sometimes resulting in a rather large file on disk. If you are dealing with a large clip with many 2D tracks over a long duration, the saved composition can reach over a gigabyte in size. Using a Camera Tracker node in a composition can make it cumbersome to load and operate. While it is possible to use the Camera Tracker node directly to composite via the 3D output, you'll achieve better performance by exporting. Once the quality of the solve is satisfactory, the Export tab can generate a "low memory" alternative by producing individual Camera 3D, Point Cloud, Ground Plane, and 3D Renderer nodes.

Before you can export the 3D scene, you must provide a bit more information about it. You'll do this using controls found in the Export tab. Cameras do not include tiltmeters, so clips do not contain metadata that indicates how the camera is tilted or oriented. This is critical information when recreating the virtual camera. It is also useful to determine the location for the center of this 3D scene. The Export tab provides various translation, rotation, and scale controls to set these options.

Unalign the 3D Scene Transforms

By default, the Export tab is set to Aligned in the 3D Scene Transform section. The Aligned setting locks the orientation and scale of the 3D scene to prevent accidentally changing it. So, before you can set the ground plane and origin location, you must change the Camera Tracker to be unaligned using the 3D Scene Transform menu in the Export tab. After you have gone through the Export settings and configured them how you want, you must set the menu back to Aligned before exporting.



Set the 3D Scene Transform menu to Unaligned before setting the ground plane

Setting the Ground Plane

The Camera Tracker has no idea if the camera is on its side or tilted in some way. So, it is up to you to indicate where the ground plane is in a clip. After choosing Unaligned from the 3D Scene Transform menu, you can begin identifying the ground plane.



Drag a selection box around marks that represent the ground

To set the ground plane, do the following:

- 1 Move to a frame with lots of green 3D locators where you can see a large part of the ground.
- 2 In the viewer, drag a selection box around the marks located on the ground in the clip.
- 3 In the Inspector Orientation section, click the Set from Selection button.

TIP: In some cases, the clip you are tracking may not have the ground in the frame. If necessary, you can set the Selection menu to XY, which indicates you are selecting points on a wall.

Setting the Origin

When it comes time to add and position new objects into your 3D scene, you can make it easier by setting the origin or center location.

To set the origin of the 3D scene, do the following:

- 1 Move to a frame that clearly shows the area you want to select as the center of the scene.
- 2 In the viewer, either select a single point or drag a selection box around a few marks located where you want to position the center of the 3D scene.
- 3 In the inspector Origin section, click the Set from Selection button.

Setting the Scale

The Camera Tracker has no idea of the size of the 3D scene, so the scale parameter is used to scale the scene output. This makes it possible to match the scale of two or more clips.

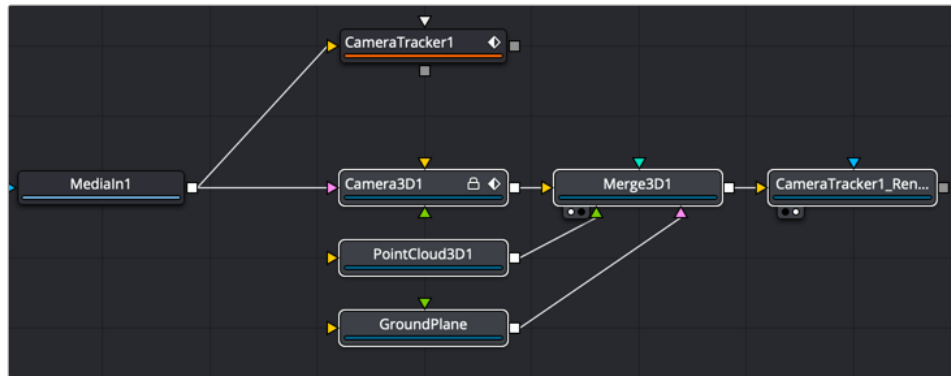
Realign the Scene

Before exporting the scene from the Camera Tracker, you must set the 3D scene Transform menu back to Aligned. Now you're ready to export.

Viewing the Exported Results

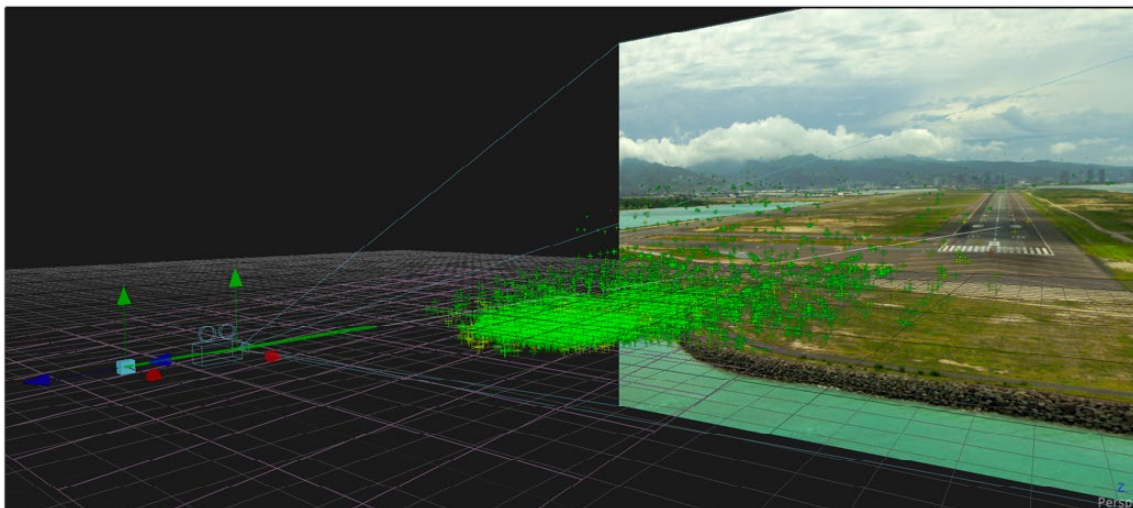
Clicking the Export button at the top of the Inspector creates a functional 3D scene with five new nodes automatically added to the node tree.

- Camera 3D
- Point Cloud
- Ground Plane
- Merge 3D
- Camera Tracker Renderer (3D Renderer)



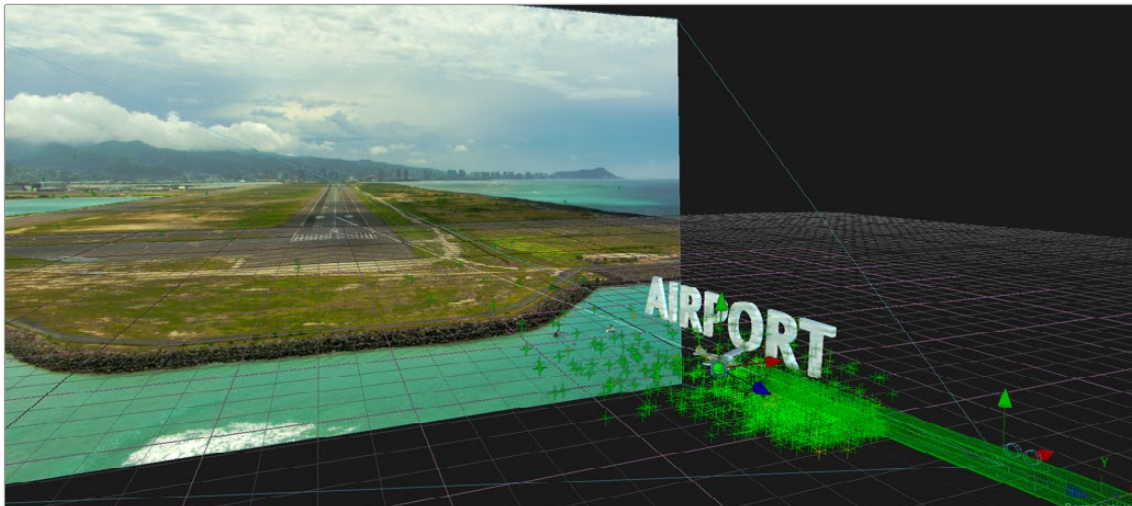
Five nodes created as a result of exporting from the Camera Tracker

To work with the 3D scene, you can select the Merge 3D and load it into one of the viewers, and then select the Camera Tracker Renderer and load that into a second viewer.



Viewing the Merge 3D shows the point cloud, ground plane, and camera

When the Merge 3D is selected, a toolbar above the viewer can add 3D test geometry like an image plane or cube to verify the precision of the 3D scene and camera. You can then connect actual 3D elements into the Merge 3D as you would any manually created 3D scene. The point cloud can help align and guide the placement of objects, and the CameraTracker Renderer is a Renderer 3D node with all the same controls.



Use the point cloud to accurately place different elements into a 3D scene

At this point, there is no need for the Camera Tracker node unless you find that you need to rerun the solver. Otherwise, you can save some memory by deleting the Camera Tracker node.